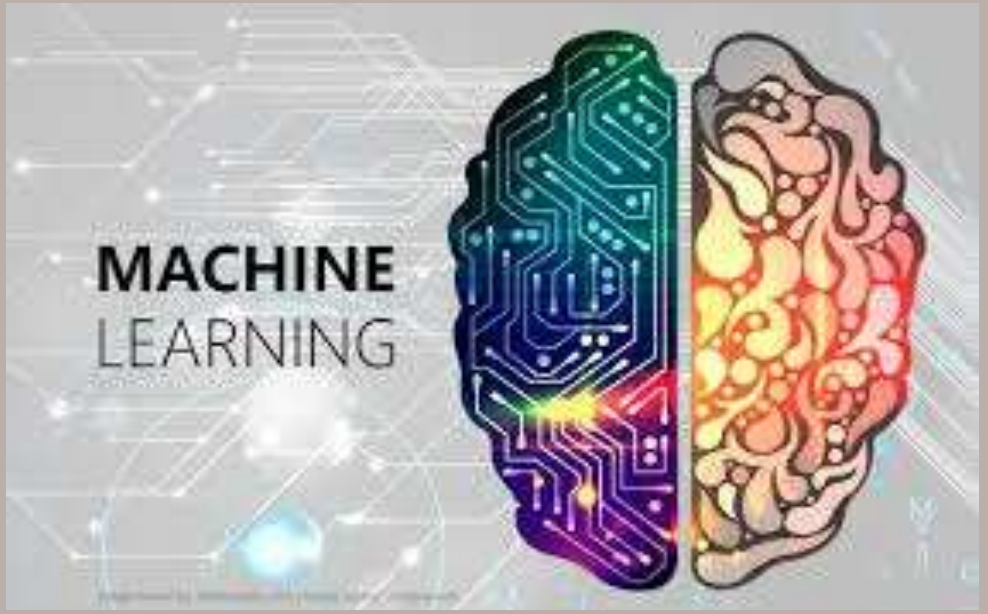
CSE 4137

Machine Learning

Module-05

Classification



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Topics: ML Classification

Week-05

- Conditional Probability
- Bayes Theorem
- Naïve Bayes SML

Logistic Regression: SML

Week-06, c-3

Introduction to probability

- Probability means possibility.
- It is a branch of mathematics that deals with the occurrence of a random event.
- Probability is a measure of the likelihood of an event to occur.



Example

- When we toss a coin, either we get Head OR Tail, only two possible outcomes are possible (H, T).
- But if we toss two coins in the air, there could be three possibilities of events to occur, such as both the coins show heads or both show tails or one shows heads and one tail.
- i.e.(H, H), (H, T),(T, T).

Probability Formula:

$$P(A) = \frac{\text{Number of favorable outcomes to A}}{\text{Total number of outcomes}}$$

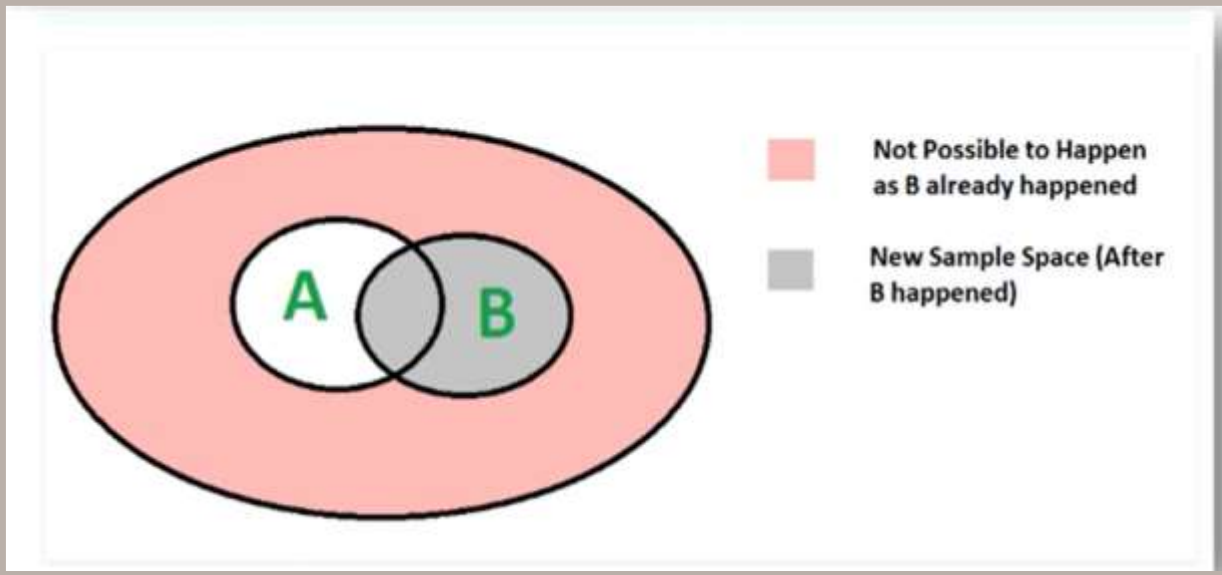
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Logistic Regression: SML

Week-06, c-3

Conditional probability

- The probability of an event occurring given that another event has already occurred is called a **conditional probability**.



Conditional Probability Formula

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Probability of *A* given *B*

Probability of *A* and *B*

Probability of *B*

Activate Windows

Logistic Regression: SML

Week-06, c-3

Conditional probability

- The probability of an event occurring given that another event has already occurred is called a **conditional probability**.

Example: Conditional probability

- Neha took two tests.
- The probability of her passing both tests is 0.6.
- The probability of her passing the first test is 0.8.
- What is the probability of her passing the second test given that she has passed the first test?

Solution:

$$P(\text{second} \mid \text{first}) = \frac{P(\text{first and second})}{P(\text{first})} = \frac{0.6}{0.8} = 0.75$$

Bayes Theorem

Week-06, c-3

Bayes Theorem

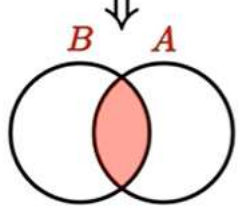
- Bayes' theorem is also known as **Bayes' rule, Bayes' law**, or **Bayesian reasoning**, which determines the probability of an event with uncertain knowledge.
- Bayes' theorem was named after the British mathematician **Thomas Bayes**.
- It is a way to **calculate the value of P(B|A) with the knowledge of P(A|B).**
- Bayes' theorem allows updating the probability prediction of an event by observing new information of the real world.

Bayes Theorem from Conditional probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$

$$P(B \cap A)$$



$$P(A \cap B) = P(B \cap A)$$

The probability that an event A will occur given another event B occurs is written as $P(A|B)$, meaning "the probability of A given B." Assuming that the probability of B is not zero, this can be calculated using the formula

From first two equations

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Bayes Theorem

Week-06, c-3

Example: Bayes Theorem

Ex You've been planning a picnic for your family. You're trying to decide whether to postpone due to rain. The chance of rain on any day is 15%. Should you postpone the picnic?

Answer: $P(\text{rain}) = 0.15$

But if we want to predict the rain more precisely then we need some more related information like morning cloud. So the problem can be redefined as follows Here Bayes formula helps to predict rain more precisely

Ex You've been planning a picnic for your family. You're trying to decide whether to postpone due to rain. The chance of rain on any day is 15%. The morning of the picnic, it's cloudy. The prob. of it being cloudy is 25% and on days where it rains, it's cloudy in the morning 80% of the time. Should you postpone the picnic?

$$P(\text{rain}) = 0.15$$
$$P(\text{cloudy}) = 0.25$$
$$P(\text{cloudy}|\text{rain}) = 0.80$$

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

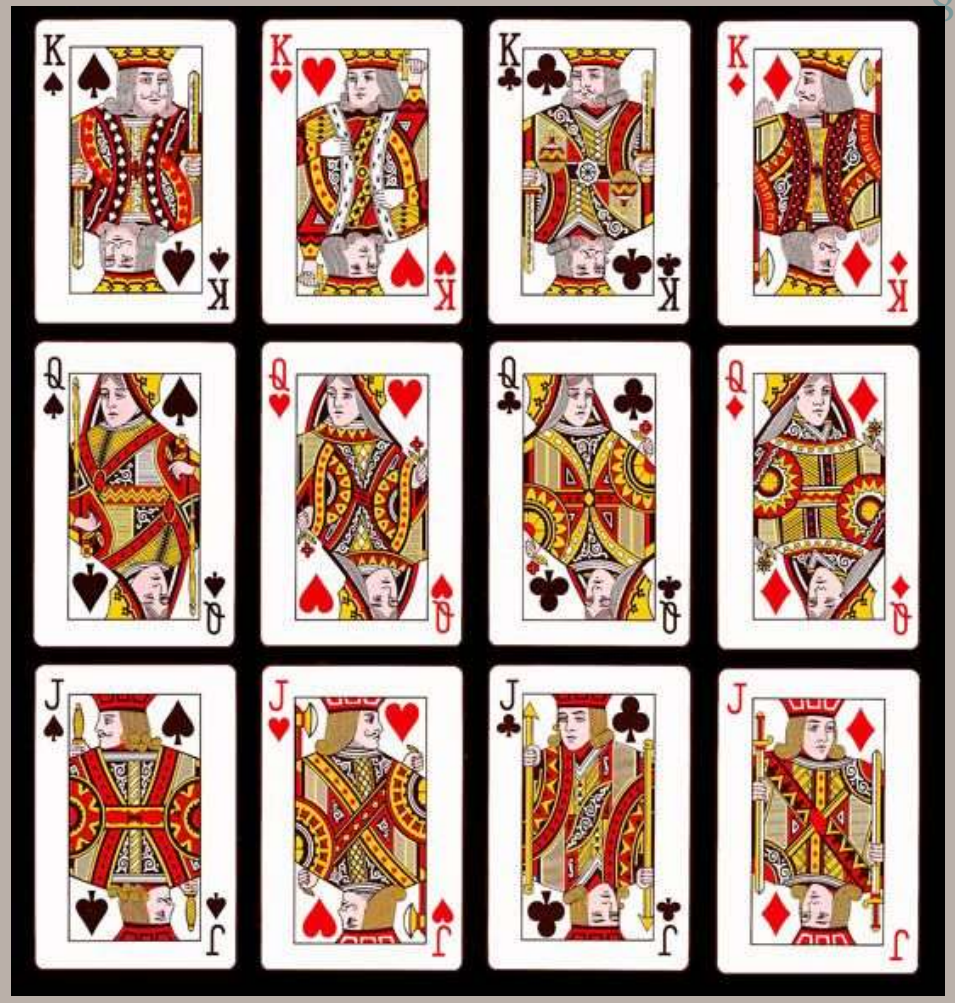
$$P(\text{rain}|\text{cloudy}) = \frac{P(\text{cloudy}|\text{rain}) \cdot P(\text{rain})}{P(\text{cloudy})}$$

$$= \frac{0.8 \cdot 0.15}{0.25}$$

$$P(\text{rain}|\text{cloudy}) = 0.48$$

The chance of rain is 3 times higher than the previous 15%.

Bayes Theorem: Example



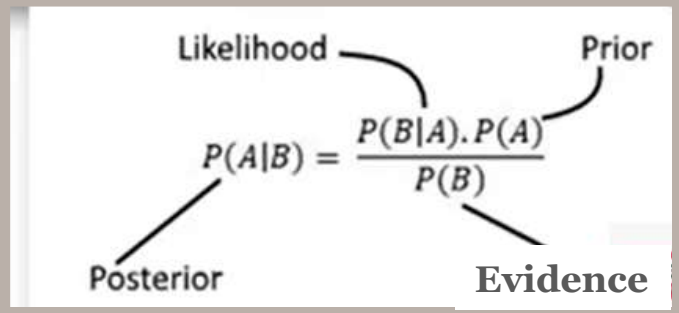
Find the P(King|face)
Here
P(King)=4/52=1/13
P(Face)=12/52=4/13
P(Face|King)=1 [if card is King then it must be Face]
According to Bayes rule
P(King|Face)=(1*1/13)/(3/13) = 1/3

Naïve Bayes Classifier

Week-06, c-3

Introduction of Naïve Bayes Classifier

- Naïve Bayes algorithm is a supervised learning algorithm, which is based on Bayes theorem and used for solving classification problems.
- It is mainly used in text classification that includes a high-dimensional training dataset.
- Naïve Bayes Classifier is one of the simple and most effective Classification algorithms which helps in building the fast machine learning models that can make quick predictions.
- It is a probabilistic classifier, which means it predicts on the basis of the probability of an object.
- Some popular examples of Naïve Bayes Algorithm are spam filtration, Sentimental analysis, and classifying articles.



Why Naïve Bayes?

➢**Naïve**:

- It is called **Naïve** because it assumes that the occurrence of a certain feature is independent of the occurrence of other features.
- Such as if the fruit is identified on the bases of color, shape, and taste, then red, spherical, and sweet fruit is recognized as an apple.
- Hence each feature individually contributes to identify that it is an apple without depending on each other.

➢**Bayes**:

- It is called Bayes because it depends on the principle of Bayes' Theorem.

Naïve Bayes Classifier

Week-06, c-3

Example: Naïve Bayes Classifier

Here, **Training data set** of "Weather" and **Corresponding Target variable** "Play".

- **Step 1:** Convert the data set into a frequency table.
- **Step 2:** Create Likelihood table by finding the probabilities.
- **Step 3:** Now, use Naive Bayesian equation to calculate the posterior probability for each class. The class with the highest posterior probability is the outcome of prediction.

Weather	Play
Sunny	No
Overcast	Yes
Rainy	Yes
Sunny	Yes
Sunny	Yes
Overcast	Yes
Rainy	No
Rainy	No
Sunny	Yes
Rainy	Yes
Sunny	No
Overcast	Yes
Overcast	Yes
Rainy	No

Frequency Table

Weather	No	Yes
Overcast		4
Rainy	3	2
Sunny	2	3
Grand Total	5	9

Likelihood table

Weather	No	Yes		
Overcast		4	=4/14	0.29
Rainy	3	2	=5/14	0.36
Sunny	2	3	=5/14	0.36
All	5	9		
	=5/14	=9/14		
	0.36	0.64		

✓**Problem:** Players will play if weather is sunny. Is this statement is correct?

- **P(Yes | Sunny) = P(Sunny | Yes) * P(Yes) / P (Sunny)**
- Here we have **P (Sunny |Yes) = 3/9 = 0.33**, **P(Sunny) = 5/14 = 0.36**, **P(Yes)= 9/14 = 0.64**
- Now, **P (Yes | Sunny) = 0.33 * 0.64 / 0.36 = 0.60**, which has higher probability.

- Here also **P(Sunny|No)=2/5=0.4**, **P(Sunny)=5/14=0.36**, **P(No)=5/14=0.36**
- **Now P(No|Sunny)=0.4*0.36/0.36=0.4**

Naïve Bayes Classifier

Week-06, c-3

Naïve Bayes Classifier with multiple features

$$P(Y/X) = \frac{P(X|Y) * P(Y)}{P(X)}$$

$$P(Y|X_1, X_2, \ldots\ldots X_n) = \frac{P(X_1|Y)*P(X_2|Y)*P(X_3|Y)\ldots\ldots P(X_n|Y)*P(Y)}{P(X_1)*P(X_2)*P(X_3)\ldots\ldots P(X_n)}$$

$$P(N/X) = \frac{P(X|N) * P(N)}{P(X)}$$

$$P(N|X_1, X_2, \ldots\ldots X_n) = \frac{P(X_1|N)*P(X_2|N)*P(X_3|N)\ldots\ldots P(X_n|N)*P(N)}{P(X_1)*P(X_2)*P(X_3)\ldots\ldots P(X_n)}$$

For predicting both Y and N values the denominators are same for both the cases so it can be ignored for the probability calculation

Prob: Predict Fever using Covid and Flu

Person	COVID (Yes/No)	Flu (Yes/No)	Fever (Yes/No)
1	Yes	No	Yes
2	No	Yes	Yes
3	Yes	Yes	Yes
4	No	No	No
5	Yes	No	Yes
6	No	No	Yes
7	Yes	No	Yes
8	Yes	No	No
9	No	Yes	Yes
10	No	Yes	No

Step-1: Prior Probability:

P(fever = yes) = 7/10

P(fever = no) = 3/10

Step-2: Conditional Probability:

	Yes	No
Covid	4/7	2/3
Flu	3/7	2/3

Given Person(flu, Covid)

P(yes| flu, Covid)= P(flu|yes)*P(covid|yes)* P(yes)

P(NO|flu, Covid)= P(flu|NO)*P(covid|No)*P(NO)

P(Yes|flu, Covid)=3/7 * 4/7 * 7/10 =0.17

P(No|flu, Covid)=2/3 * 2/3 * 3/10 =0.13

So a patient carrying Covid and Flu the model predicts Fever

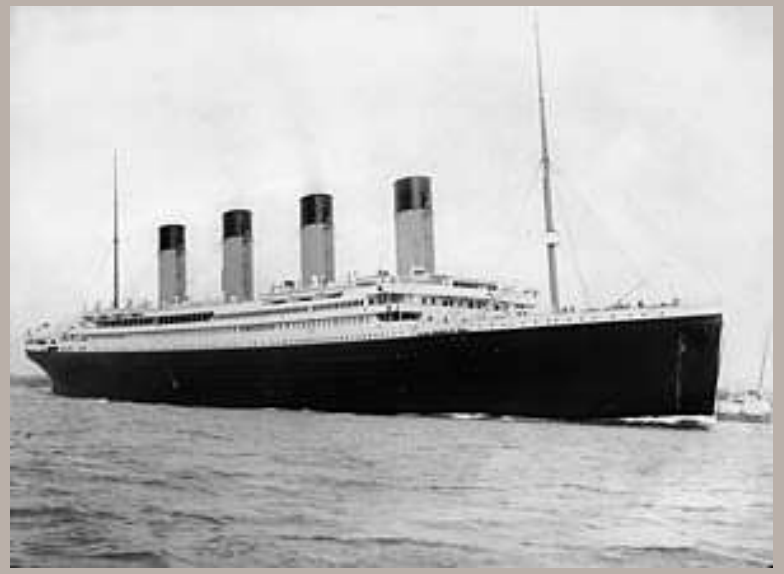
Example: Naïve Bayes Classification

Naïve Bayes Classifier : SML

Week-05, c-3

Example: Prediction of survival of Titatic

RMS **Titanic** was a British ocean liner that sank on 15 April 1912 as a result of striking an iceberg on her maiden voyage from Southampton, England, to New York City, United States. Of the estimated 2,224 passengers and crew aboard, approximately 1,500 died (figures vary), making the incident one of the deadliest peacetime sinkings of a single ship



According to Titanic's general arrangement plans, the ship could accommodate 833 First Class Passengers, 614 in Second Class and 1,006 in Third Class, for a total passenger capacity of 2,453. In addition, Titanic's capacity for crew members exceeded 900, as most documents of the original configuration have stated that the full carrying capacity for passengers and crew was approximately 3,547.



Naïve Bayes Classifier : SML

Week-05, c-3

Titanic: Dataset Description

Variable	Definition	Key
survival	Survival	0 = No, 1 = Yes
pclass	Ticket class	1 = 1st, 2 = 2nd, 3 = 3rd
sex	Sex	
Age	Age in years	
sibsp	# of siblings / spouses aboard the Titanic	
parch	# of parents / children aboard the Titanic	
ticket	Ticket number	
fare	Passenger fare	
cabin	Cabin number	
embarked	Port of Embarkation	C = Cherbourg, Q = Queenstown, S = Southampton

model.fit(inputs_n_target)

Naïve Bayes Classifier : SML

Week-05, c-3

Example: Prediction of survival of Titatic

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.metrics import
confusion_matrix,classification_report
from sklearn.model_selection import
train_test_split
from sklearn.naive_bayes import GaussianNB
```

```
df=pd.read_csv('F:\\python\\titanic.csv')
```

```
df.drop(['PassengerId','Name','SibSp','Parch','Ticket
','Cabin','Embarked'],axis='columns',inplace=True)
```

```
n=len(df[(df['Survived']==0) & (df['Sex']=='female')])
t=len(df[df['Survived']==0])
p=n/t*100
```

```
df.isnull().sum()
```

```
df['Age'].fillna(value=df['Age'].mean() ,
inplace=True)
```

```
df['Sex'].replace({'male':1,'female':0},inplace
=True)
```

```
X = df.drop('Survived',axis='columns')
y = df.Survived
```

```
X_train, X_test, y_train, y_test =
train_test_split(X,y,test_size=0.3)
```

```
model = GaussianNB()
model.fit(X_train,y_train)
```

```
y_pred=model.predict(X_test)
```

```
cm=confusion_matrix(y_test,y_pred)
```

```
print(classification_report(y_test,y_pred))
```

Naïve Bayes Classifier : SML

Week-05, c-3

Example: Contd.

Performance of the Model

```
cm=confusion_matrix(y_test,y_pred)
```

```
cm
```

```
array([[127,  35],
       [ 30,  76]], dtype=int64)
```

```
print(classification_report(y_test,y_pred))
```

```
              precision    recall  f1-score   support

           0       0.81      0.78      0.80       162
           1       0.68      0.72      0.70       106

    accuracy                           0.76       268
   macro avg       0.75      0.75      0.75       268
weighted avg       0.76      0.76      0.76       268
```

Test the Model with new input

```
a=[[1,0,18,227]]
b=model.predict(a)
if(b[0]==1):
    print('Survived')
else:
    print('Not Survived')
```

End of Chapter